Digital Audio-Video Signal Processing

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1 Question - 10 pts

Draw the block diagram of a DPCM codec. Consider adopting the following 1-bit scalar quantizer

$$y = +Q \quad \text{if } x \geq 0 \tag{1}$$
$$y = -Q \quad \text{if } x < 0 \tag{2}$$

Describe the output of the DPCM decoder in two cases (hint: generate a sample input signal and compute the output):

- A constant signal
- A signal rapidly increasing or decreasing

For each case, suggest how Q should be tuned. Briefly propose an adaptive scheme that can handle both scenarios.

2 Question - 5 pts

Sketch the block diagram of a perceptual audio codec, illustrating the functionality of the individual blocks. Describe in detail the perceptual model.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a discrete memoryless source with finite alphabet, and $D(R)$ the corresponding rate-distortion curve
   - [T] [F] $D(0) = H(X)$
   - [T] [F] If $\langle D, R \rangle$ is achievable, than $\langle D, R + \Delta R \rangle$, $\Delta R > 0$ is achievable
   - [T] [F] $R(0) = \infty$
   - [T] [F] $H(X) \leq \log_2(M)$, where M is the alphabet size

2. Consider a speech codec
   - [T] [F] The perceptual model determines which formants needs to be encoded
   - [T] [F] Direct quantization of reflection coefficients might cause instability
   - [T] [F] The long-time predictor is used to describe the spectral envelope of the signal
3. Consider a video coding system
   - [T] [F] The bit-rate needed to encode each frame of a sequence is always constant
   - [T] [F] For a target distortion level, the bitrate depends on the motion estimation algorithm adopted
   - [T] [F] Block-based motion estimation typically employs partially overlapping blocks
   - [T] [F] In a block-based motion estimation algorithm, the PSNR of the displaced frame difference increases if the block size increases

4 Question - 12 pts

Load the data by means of the Matlab command `load tracks.txt`. The columns of the matrix `tracks` contain the following data

1. Heart beat rate (beats/minute)
2. Speed (Km/h $\times$ 10)
3. Cadence (to be ignored)
4. Elevation (meters)

The sampling time is $T = 5$ sec. Pre-process the data as follows:

- Filter the data with a median filter (`medfilt1`) to remove the outliers
- Convert speed data in meters/sec
- Compute the distance covered at each time instant
- Compute the instantaneous gradient (elevation gain / distance)

Data needs to be coded for storage.

- Design a scalar uniform quantizer to encode the following two sources independently: heart beat rate and gradient. Define both the granular region and the quantization step size.
- Compute the number of bits / sample needed to encode the two sources with and without entropy coding of the quantization indexes.
- Plot the quantized gradient vs. heart beat rate in a scatter plot. Can you envisage bit-rate savings by employing a joint encoding scheme?